

Theory Week 12

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Mechanics 3 CAB G59
WhatsApp group



- Find all doc's on my website (link in WhatsApp bio)
- remember to do the STACK
- Study Center
→ Wednesday 18:00-20:00 (ETC 4)

12	02/12/24	single-DOF forced vibrations	4.3.3
	04/12/24	multi-DOF free vibrations	4.4-4.4.2
exercise 12: single-DOF vibrations			

4.3.3 Forced vibrations

general solutions for forced vibrations: already know this for free vibrations

$$f(t) = \hat{f} \cos(\Omega t) \Rightarrow x(t) = x_{\text{hom}}(t) + \frac{|\hat{f}|}{\omega_0^2} V \cos(\Omega t - \varphi)$$

excitation freq.

$$f(t) = \hat{f} \sin(\Omega t) \Rightarrow x(t) = x_{\text{hom}}(t) + \frac{|\hat{f}|}{\omega_0^2} V \sin(\Omega t - \varphi)$$

with magnification and phase delay

$$V = \frac{1}{\sqrt{(1-\eta^2)^2 + 4D^2\eta^2}}, \quad \varphi = \begin{cases} \arctan\left(\frac{2D\eta}{1-\eta^2}\right), & \text{if } \eta \leq 1 \\ \pi + \arctan\left(\frac{2D\eta}{1-\eta^2}\right), & \text{if } \eta > 1 \end{cases}$$

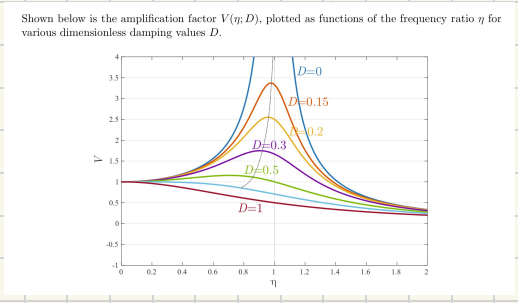
magnification factor phase

freq. ratio
where $\eta = \frac{\Omega}{\omega_0}$

again we have given the solns
→ we just use them

steady state: $x(t) \approx x_{\text{part}}(t)$ as $t \rightarrow \infty$

resonance: $\Omega = \omega_0$, i.e. when the system is driven at natural frequency
→ amplitude peaks



Multi-DOF vibrations

Two alternatives (shown in Lecture)

① derive non-linear EOM (Lagrange / LMB / AMB), then linearize about equilibrium.

$$F(q, \dot{q}, \ddot{q}) = F^{\text{ext}}(t)$$

equations of motion for multiple-DOF vibrations:

$$M\ddot{x} + C\dot{x} + Kx = F(t)$$

with the three constant matrices¹⁹

$$M = \frac{\partial F}{\partial \ddot{q}}(q_0, 0, 0), \quad C = \frac{\partial F}{\partial \dot{q}}(q_0, 0, 0), \quad K = \frac{\partial F}{\partial q}(q_0, 0, 0),$$

M = mass matrix // C = damping matrix // K = stiffness matrix

→ We assume $Q^{\text{nc}} = -C\dot{q}$, $C \in \mathbb{R} \Rightarrow C$ & K are sym., positive semi-definite, constant $\lambda_i \geq 0$

② expand kinetic & potential energies about equilibrium to quadratic order & then to obtain linearized equations from the Lagrange eq.

kinetic and potential energy for multiple-DOF linear(ized) vibrations:

$$T = \frac{1}{2} \dot{x} \cdot M \dot{x}, \quad V = \frac{1}{2} x \cdot K x$$

Resulting EOM same as above:

equations of motion for multiple-DOF vibrations:

$$M \ddot{x} + C \dot{x} + K x = F(t)$$

linearized system matrices around a stable equilibrium q_0 :

$$M = \frac{\partial^2 T}{\partial \dot{q} \partial \dot{q}}(q_0, 0), \quad C = -\frac{\partial Q^{nc}}{\partial \dot{q}}(q_0, 0), \quad K = \frac{\partial^2 V}{\partial q \partial q}(q_0) - \frac{\partial Q^{nc}}{\partial q}(q_0, 0)$$

Free undamped vibrations

$$\underline{F^{ext}} = \underline{0}, \quad C = 0 \quad \Rightarrow \quad \underline{M} \ddot{x} + \underline{K} x = 0$$

Eigenvalue problem for eigenfrequencies & eigenmodes

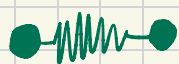
$$\begin{cases} (\underline{K} - \omega_j^2 \underline{M}) \hat{x}^* = 0 \\ \det(\underline{K} - \omega_j^2 \underline{M}) = 0 \end{cases} \quad \omega_j = \text{eigenfrequencies}$$

- zero eigenfrequencies ($\lambda_i = 0 \rightarrow \omega_j = 0$)

Particle system moves like a rigid body (particles distance unchanged)

$\Rightarrow$ rigid body motion

$\Rightarrow$ Every system has as many pairs of zero eigenfreq. as it has indep. rigid body modes.



in 1D $\rightarrow$ 1 rigid body mode

in 2D $\rightarrow$ 3 rigid body modes

General solns for $n-l \leq n$ rigid body modes & l eigenfreq. $\neq 0$

(& ofc. free & undamped)

eigenfrequencies ω_j and eigenmodes $\hat{x}_j$ are obtained from

$$\det(-\omega_j^2 M + K) = 0 \quad \text{and} \quad (-\omega_j^2 M + K) \hat{x}_j = 0$$

general solution of **free vibrations** for multiple-DOF systems:

$$x(t) = \sum_{j=1}^n x_j(t) = \sum_{j=1}^l c_j \cos(\omega_j t + \varphi_j) \hat{x}_j + \sum_{j=l+1}^n (a_j + b_j t) \hat{x}_j$$

$$a_j, b_j, c_j \in \mathbb{R}$$